RETAIL INVENTORY AND STOCK MANAGEMENT SYSTEM

Team: Team E10

Course: Software Engineering Mini-Project – Deliverables Part 1

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional, verifiable functional, non-functional and security requirements of the Retail Inventory and Stock Management System. The SRS is intended for the development team, testers, course instructor and future maintainers of the system.

The system is a menu-driven C/C++ application designed to support basic inventory operations for small retail stores. It provides facilities for maintaining product records, searching and viewing products, adjusting stock quantities, checking stock availability, viewing stock movement history and generating a basic inventory summary.

1.2 Scope

The Retail Inventory and Stock Management System is a C/C++ application designed to organize basic inventory operations in a small retail environment. The system allows users to add, modify, remove, search and view product records, adjust stock quantities, check available stock, maintain stock movement history and view inventory information.

The system shall store product records and stock movement history using file handling and shall load previously stored data when the application starts.

In scope: product record management, product search and viewing, stock quantity adjustment, stock availability checking, stock movement history, inventory listing, inventory summary, low-stock identification and file-based data persistence.

The system does not require an external database or hosted storage and is intended to operate as a local C/C++ application.

In scope: product management, product search, product updates, product removal, stock adjustment, stock availability checking, stock movement history, user authentication and file-based persistence.

Out of scope: graphical or web interface, online shopping, payment processing, supplier or purchase-order management, external database integration, networking, barcode hardware and integration with external retail systems.

1.3 Definitions, Acronyms and Abbreviations

Term	**Meaning**

Term	Meaning
SRS	Software Requirements Specification
FR	Functional Requirement
VFR	Verifiable Functional Requirement
RTM	Requirements Traceability Matrix
C/C++	Programming languages used to implement the system
Product Record	Stored information describing a product, including its identification and inventory details
Stock Movement	An increase or decrease in the quantity of a product
Low-Stock Item	A product whose available quantity is at or below the configured low-stock threshold
Inventory User	User who performs product and inventory operations in the system

1.4 References

• IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.

• ISO/IEC/IEEE 29148:2018, Systems and software engineering - Requirements engineering.

• Course document: Software Engineering Mini-Project Deliverables Part-1.

1.5 Overview

Section 2 provides the overall description of the Retail Inventory and Stock Management System, including its product perspective, functions, users, operating environment, constraints, assumptions and dependencies.

Section 3 specifies the detailed system requirements, including external interfaces, use cases, functional requirements, verifiable functional requirements, non-functional requirements, business rules and data requirements.

Section 4 describes the security objectives, assets, threats and security requirements of the system.

Section 5 provides the Requirements Traceability Matrix linking system requirements with their corresponding use cases and verification activities.

2. Overall Description

2.1 Product Perspective

The Retail Inventory and Stock Management System is a standalone, menu-driven C/C++ application intended to support basic inventory operations for small retail stores. The system provides functions for maintaining product records, searching and viewing products, adjusting stock quantities, checking stock availability, viewing stock movement history and generating a basic inventory summary.

The system uses file handling for persistent storage of product records and stock movement history. It does not require an external database or hosted storage.

2.2 Product Functions

The system shall provide the following functions:

- Allow an authorized user to enter the inventory application.
- Add new product records with product ID, name, category, unit price and quantity.
- Modify existing product records.
- Remove existing product records.
- Search for products using product ID or name.
- Display details of a selected product.
- Increase or decrease product stock quantity.
- Display the current available quantity of a selected product.
- Record stock adjustments with product ID, quantity change, date/time and adjustment type.
- View stock movement history for a product.
- Display a list of all stored product records.
- Generate a basic inventory summary showing product quantities and low-stock items.
- Store product records and stock movement history in files.
- Load previously stored product and stock history data when the application starts.

2.3 User Classes and Characteristics

Actor	**Description**
Administrator	Performs administrative inventory operations, including accessing the application, managing product records, viewing product information, checking stock availability, viewing stock movement

Actor	Description
	history and generating inventory information.
Inventory Staff	Performs day-to-day inventory operations, including managing product records, searching and viewing products, adjusting stock quantities, checking stock availability and viewing stock movement history.

2.4 Operating Environment

The system shall operate on a computer capable of running a C/C++ application. The application shall use a console-based interface and standard file input/output operations for persistent storage. The system shall use a keyboard for input and a display for output.

2.5 Design and Implementation Constraints

- The system shall be implemented using C or C++.
- The system shall use file handling for persistent storage.
- The system shall use a menu-driven console interface.
- The system shall not require an external database or hosted storage.
- Product IDs shall uniquely identify stored products.
- Product quantity shall not be negative.
- A configurable threshold shall be used to identify low-stock products.

2.6 Assumptions and Dependencies

- The target computer shall have a compatible C/C++ compiler or executable environment.
- The application shall have access to the files used for storing product and stock movement data.
- Users shall provide valid product information when performing inventory operations.
- The configured low-stock threshold shall be available to the system when generating the inventory summary.
- The system shall be able to access previously stored files when loading inventory data at startup.

3. Specific Requirements

3.1.1 User Interfaces

The system shall provide a menu-driven console interface for inventory operations. The interface shall provide options for entering the application, managing product records, searching and viewing products, adjusting stock quantities, checking stock availability, viewing stock movement history, viewing inventory records and generating a basic inventory summary.

The system shall display clear prompts for user input and appropriate messages for successful operations, invalid input and unsuccessful searches.

3.1.2 Hardware Interfaces

The system shall use a standard computer keyboard for user input and a display for console output. No specialized hardware interface is required.

3.1.3 Software Interfaces

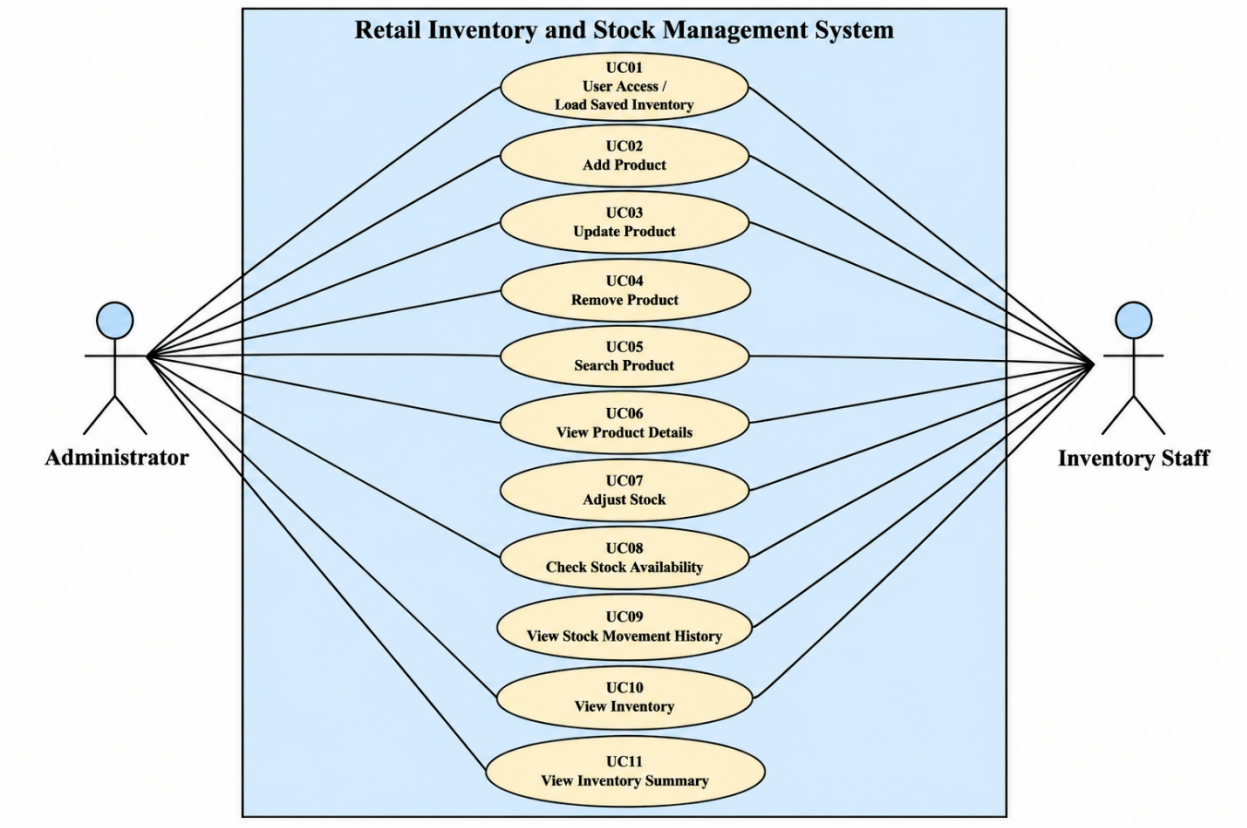
The system shall be implemented using C or C++ and shall use standard file input/output operations for persistent storage. Product records and stock movement history shall be stored and retrieved using local files.

3.1.4 Communication Interface

The system shall operate locally and shall not require network communication or external communication services.

3.2 Use Case Model

The use case model describes the interactions between the Administrator, Inventory Staff and the Retail Inventory and Stock Management System. The use cases represent the main inventory operations performed by the system users.



3.2.1 Use Case List by Actor

Use Case ID	Use Case
UC01	User Access / Load Saved Inventory
UC02	Add Product
UC03	Update Product
UC04	Remove Product
UC05	Search Product
UC06	View Product Details
UC07	Adjust Stock

Use Case ID	Use Case
UC08	Check Stock Availability
UC09	View Stock Movement History
UC10	View Inventory
UC11	View Inventory Summary

3.2.2 Use Case Descriptions

UC01 – User Access / Load Saved Inventory
Actor: Administrator / Inventory

Description: The user enters the inventory application and the system loads previously stored product records and stock movement history.

Precondition: The application is available and the required data files are accessible.

Main Flow:

1. User starts the application.
2. System provides access to the inventory application.
3. System loads previously stored product records.
4. System loads previously stored stock movement history.
5. System displays the inventory menu.

Postcondition: The user can access the inventory functions and previously stored data is available.

UC02 – Add Product
Actor: Administrator / Inventory Staff

Description: The user adds a new product record containing product ID, name, category, unit price and quantity.

Precondition: The user has entered the inventory application.

Main Flow:

1. User selects Add Product.

2. User enters the product ID, name, category, unit price and quantity.

3. System validates the product information.

4. System checks that the product ID is unique.

5. System stores the new product record.

Alternative Flow: If the product ID already exists or the required information is invalid, the system rejects the product record

Postcondition: A valid product record is stored.

UC03 – Update Product
Actor: Administrator / Inventory Staff

Description: The user modifies the details of an existing product record.

Precondition: The user has entered the inventory application and the product exists.

Main Flow:

1. User selects Update Product.

2. User identifies the product.

3. System retrieves the product record.

4. User enters the modified product information.

5. System validates the information.

6. System stores the updated product record.

Postcondition: The product record contains the updated information.

UC04 – Remove Product
Actor: Administrator

Description: The user removes an existing product record from the inventory.

Precondition: The user has entered the inventory application and the product exists.

Main Flow:

1. User selects Remove Product.

2. User identifies the product.

3. System retrieves the product record.

4. User confirms the removal.

5. System removes the product record.

Postcondition: The selected product record is removed from the stored inventory.

UC05 – Search Product
Actor: Administrator / Inventory Staff

Description: The user searches for a product using its product ID or name.

Precondition: The user has entered the inventory application.

Main Flow:

1. User selects Search Product.

2. User enters a product ID or product name.

3. System searches the stored product records.

4. System displays the matching product record.

Alternative Flow: If no product matches the search key, the system displays a not-found message.

Postcondition: The matching product information is displayed or the user is informed that no match exists.

UC06 – View Product Details
Actor: Administrator / Inventory Staff

Description: The user views the details of a selected product.

Precondition: The selected product exists.

Main Flow:

1. User selects a product.

2. System retrieves the stored product record.

3. System displays the product details.

Postcondition: The selected product details are displayed.

UC07 – Adjust Stock
Actor: Inventory Staff

Description: The user increases or decreases the stock quantity of a product.

Precondition: The product exists and the user has entered the inventory application.

Main Flow:

1. User selects Adjust Stock.

2. User identifies the product.

3. User enters the stock adjustment and adjustment type.

4. System validates the stock adjustment.

5. System updates the product quantity.

6. System records the stock adjustment with product ID, quantity change, date/time and adjustment type.

Alternative Flow: If a stock-out quantity is greater than the available quantity, the system rejects the operation.

Postcondition: The product quantity is updated and the successful stock adjustment is recorded.

UC08 – Check Stock Availability
Actor: Administrator / Inventory Staff

Description: The user checks the current available quantity of a selected product.

Precondition: The selected product exists.

Main Flow:

1. User selects Check Stock Availability.

2. User identifies the product.

3. System retrieves the current quantity.

4. System displays the available quantity.

Postcondition: The current stock quantity is displayed.

UC09 – View Stock Movement History
Actor: Inventory Staff

Description: The user views the recorded stock movements for a product.

Precondition: The product and its stock movement history are available.

Main Flow:

1. User selects View Stock Movement History.

2. User identifies the product.

3. System retrieves the recorded stock movements.

4. System displays the stock movement history.

Postcondition: The stock movement history for the selected product is displayed.

UC10 – View Inventory
Actor: Administrator / Inventory Staff

Description: The user views a list of all stored product records.

Precondition: The user has entered the inventory application.

Main Flow:

1. User selects View Inventory.

2. System retrieves the stored product records.

3. System displays the product records.

Postcondition: The stored inventory list is displayed.

UC11 – View Inventory Summary
Actor: Administrator

Description: The user views a basic inventory summary containing product quantities and low-stock items.

Precondition: Product records are available and the low-stock threshold is configured.

Main Flow:

1. User selects View Inventory Summary.

2. System retrieves the stored product quantities.

3. System compares available quantities with the configured low-stock threshold.

4. System identifies low-stock products.

5. System displays the inventory summary.

Postcondition: The inventory summary and low-stock items are displayed.

3.3 Functional Requirements

Requirement ID	Functional Requirement	Description
FR01	User Access	The system shall allow an authorized user to enter the inventory application.
FR02	Add Product	The system shall allow users to add a new product record with a unique product ID, name, category, unit price and quantity.
FR03	Modify Product	The system shall allow users to modify the details of an existing product.
FR04	Remove Product	The system shall allow users to remove an existing product record.
FR05	Search Product	The system shall allow users to search for a product using the product ID or product name.
FR06	View Product Details	The system shall display the details of a selected product.
FR07	Adjust Stock	The system shall allow users to increase or decrease the stock quantity of a product.
FR08	Check Stock Availability	The system shall display the current available quantity for a selected product.
FR09	Record Stock Adjustments	The system shall maintain a record of stock adjustments including product ID, quantity change, date/time and adjustment type.
FR10	View Stock Movement History	The system shall allow users to view the stock movement history for a product.
FR11	View Inventory	The system shall display a list of all stored product records.
FR12	Generate Inventory Summary	The system shall generate a basic inventory summary showing product quantities and low-stock

Requirement ID	Functional Requirement	Description
		items.
FR13	Store Inventory Data	The system shall store product records and stock movement history in files.
FR14	Load Stored Data	The system shall load previously stored product and stock history data when the application starts.

3.4 Non-Functional Requirements

Requirement ID	Category	Requirement
NFR01	Usability	The system shall provide a simple menu-driven console interface with clear prompts and messages for inventory operations.
NFR02	Input Validation	The system shall provide appropriate validation messages when users enter invalid or incomplete product or stock information.
NFR03	Reliability	The system shall maintain accurate product and stock information during inventory operations.
NFR04	Data Persistence	The system shall preserve product records and stock movement history using file-based storage so that stored information remains available after the application is closed and restarted.
NFR05	Maintainability	The system shall be organized into modules for product record management, stock adjustment, inventory lookup and history/report viewing to support implementation and testing.
NFR06	Portability	The system shall be implementable using C or C++ and standard file input/output operations on a compatible computer environment.
NFR07	Performance	The system shall perform basic inventory operations such as creation, search, update,

Requirement ID	Category	Requirement
		deletion, quantity adjustment and data retrieval without unnecessary processing delays under normal usage.
NFR08	Local Operation	The system shall operate as a local application and shall not depend on an external database, hosted storage or network communication for its basic inventory operations.

3.5 Business Rules

Rule ID	Business Rule	Description
BR01	Unique Product ID	Each product shall have a unique product ID.
BR02	Required Product Information	A product record shall contain product ID, name, category, unit price and quantity.
BR03	Non-Negative Values	Unit price and quantity shall not be negative.
BR04	Valid Product Search	A search using a valid stored product ID shall return the matching product record.
BR05	Unsuccessful Search	If a search key does not match any stored product, the system shall display a not-found message.
BR06	Stock-In Calculation	A valid stock-in operation shall increase the previous quantity by the entered amount.
BR07	Stock-Out Restriction	A stock-out operation shall not be allowed when the requested quantity is greater than the available quantity.

Rule ID	Business Rule	Description
BR08	Stock Adjustment History	Every successful stock adjustment shall be recorded with the product ID, quantity change, date/time and adjustment type.
BR09	Data Persistence	Product records and stock movement history shall remain available after the application is closed and restarted.
BR10	Low-Stock Threshold	A product shall be identified as low-stock when its available quantity is at or below the configured threshold.

3.6 Data Requirements

Data ID	Data Entity	Required Data	Storage Requirement
DR01	Product Record	Product ID, name, category, unit price and quantity	Shall be stored in a local file.
DR02	Product Identification	Product ID	Shall uniquely identify each stored product.
DR03	Stock Quantity	Current available quantity of a product	Shall be maintained and updated when stock is increased or decreased.
DR04	Stock Movement History	Product ID, quantity change, date/time and adjustment type	Shall be stored in a local history file and retained after application restart.

4. Security

4.1 Security Objectives

SO-01: Ensure that only authorized users can access the inventory application and perform inventory operations.

SO-02: Protect product records and stock information from unauthorized modification or removal.

SO-03: Maintain the integrity of stock quantities and stock movement history during inventory operations.

SO-04: Ensure that stored inventory data remains available and consistent after application restart.

4.2 Assets and Threats

Asset	Threat	Protection
User Access	Unauthorized access to inventory functions	Access restricted to authorized users
Product Records	Unauthorized modification or removal	Controlled access and input validation
Stock Records	Invalid or unauthorized stock changes	Stock adjustment validation and controlled updates
Stock Movement History	Unauthorized modification or loss of history data	Controlled file access and file-based persistence

4.3 Security Requirements

ID	Security Requirement
SEC-01	The system shall allow inventory functions to be accessed only by authorized users.
SEC-02	The system shall restrict unauthorized modification or removal of product records.
SEC-03	The system shall validate stock adjustments before updating stored stock quantities.
SEC-04	The system shall maintain the integrity of stock movement history when successful stock adjustments are recorded.

5 — Requirements Traceability Matrix

Requirement ID	Requirement	Related Use Case	Verification
FR01	User can enter the application.	UC01 – User Access / Load Saved Inventory	Access control testing
FR02	User can add product records.	UC02 – Add Product	Product creation testing
FR03	User can modify product details.	UC03 – Update Product	Product update testing
FR04	User can remove product records.	UC04 – Remove Product	Product deletion testing
FR05	User can search by product ID or name.	UC05 – Search Product	Search testing
FR06	System displays selected product details.	UC06 – View Product Details	Information display testing
FR07	User can increase or decrease stock.	UC07 – Adjust Stock	Quantity adjustment testing
FR08	System displays current stock quantity.	UC08 – Check Stock Availability	Stock lookup testing
FR09	System records stock adjustments.	UC07 – Adjust Stock	History logging testing
FR10	User can view product stock history.	UC09 – View Stock Movement History	History retrieval testing
FR11	System lists all product records.	UC10 – View Inventory	Inventory listing testing
FR12	System provides a basic stock summary.	UC11 – View Inventory Summary	Summary generation testing

Requirement ID	Requirement	Related Use Case	Verification
FR13	System stores data in files.	UC02 – Add Product / UC07 – Adjust Stock	File persistence testing
FR14	System loads saved data on startup.	UC01 – User Access / Load Saved Inventory	Data loading testing

5.2 Verifiable Functional Requirements Traceability

Requirement ID	Requirement	Related Use Case	Verification
VFR01	Duplicate product IDs are rejected.	UC02 – Add Product	Duplicate ID validation
VFR02	Invalid or incomplete product input is rejected.	UC02 – Add Product	Input validation testing
VFR03	Valid product search returns the matching record.	UC05 – Search Product	Search result verification
VFR04	Unknown search key returns a not-found message.	UC05 – Search Product	Negative search testing
VFR05	Stock-in updates quantity by the entered amount.	UC07 – Adjust Stock	Stock-in calculation test
VFR06	Stock-out cannot exceed available quantity.	UC07 – Adjust Stock	Insufficient stock validation
VFR07	Successful stock changes are logged.	UC07 – Adjust Stock	History logging verification
VFR08	Saved records remain after restart.	UC01 – User Access / Load Saved Inventory	Persistence/restart test
VFR09	Low-stock items meet the configured threshold.	UC11 – View Inventory Summary	Threshold validation test